

Sawan H N

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology, Kanpur	7.43/10
2021	TSBIE(XII)	Narayana Junior College, Hyderabad	98.6%
2019	CBSE(X)	Jawahar Navodaya Vidyalaya,Davanagere	95.6%

Key Projects

Federated Learning Based Content Caching in Edge Computing| *Prof. Rajesh Hegde, IIT Kanpur*🔗(May'23-Ongoing)

- Employed **Federated Learning** to develop a privacy-preserving content caching scheme where users do not have to share their private data to the central server but train locally on their data and share the model parameters with the server
- Implemented caching algorithms like **random**, **Most frequently used**, **m-epsilon greedy** and **thompson sampling**
- Explored the **stacked Autoencoder** which trains an one-hidden layer **neural network** to learn the hidden features of the training data and implemented it on the MovieLens dataset to predict the most popular movies among the users

Mess Automation System IITK| *Course project under Prof.Indranil Saha, CSE Dept, IIT Kanpur*🔗(Jan'23-Apr'23)

- Worked with a **9 membered** team to develop a **web application** to automate various activities of a hostel mess at IITK
- Contributed in development of features like **meal booking**, **menu updation**, **rebate application** and **ticket verification**
- Utilised **EJS**, **CSS**, **JavaScript** for frontend, **Node.js** for backend, **MongoDB** for database, and collaborated via **Github**
- Contributed in documentation of Requirement Document, Design Document, User Manual and Test Document

CSE Bubble|*Course project under Prof.Urbi Chatterjee, CSE Dept, IIT Kanpur*🔗(Jan'23-Apr'23)

- Built **Verilog HDL** processor with a custom Instruction Set Architecture, having word and architecture size of 32 bits
- Designed an **ALU** and **FSM** to control the processor's signals for fetching, decoding, and executing instructions using ALU
- Developed MIPS code for **Bubble Sort**, translated it to **Machine Code** and tested the processor with different inputs

Application of Reinforcement Learning|*Association of Computing Activities, IIT Kanpur*🔗(May'22- Jul'22)

- Learned Python libraries, **Markov Decision process**, policy iteration, value iteration and Model free control/prediction
- Built ML models for price estimation using multi-variable **linear regression** and classification using **logistic regression**
- Implemented RL models for **frozen lake problem** and **Q-learning** algorithms for the **cartpole** and **taxi problems**

Space Travel System|*Course project under Prof.Indranil Saha, CSE Dept, IIT Kanpur*🔗(Mar'23- Mar'23)

- Developed a **Space Travel System** using **C++** and the concepts of **Object-Oriented Programming** for implementation
- Developed three distinct **classes** (Passenger, Astronaut, Commander) and features like **booking**, **deletion** and **updation**
- Built a STS that manages assignment of travellers and validates tickets based on distance, travel date and passenger details

Technical Skills

Programming Languages	Libraries	Softwares and Tools
C, C++, Python, Verilog HDL, JavaScript	Numpy, Pandas, Matplotlib, TensorFlow	Node.js, MongoDB, Git, L ^A T _E X

Relevant Courses

(*Ongoing)

Data Structure and Algorithms	Fundamentals of computing	Discrete Mathematics
Probability for Computer Science	Introduction to Electronics	Logic in Computer Science
Software developement and operations	Linear Algebra and differential equations	Computer Architecture

Positions of Responsibility

Secretary, Games and Sports Council, IITK(May'22-Apr'23)

- Contributed to successful organisation of volleyball tournaments in **Freshers'Inferno'23**, **Inferno'23** and **IVL'23**
- Actively participated in **registration** management, **logistic** coordination, also served as a **refree** for numerous matches

Secretary, Musicals Events-Antragni'22(Jul'22-Mar'23)

- Coordinated with a team of over **20 organisers**, **secretaries** and **volunteers** for conduction of **6 events in 3 days**
- Managed event registration and technical requirements, ensuring coordination with **judges** for successful event conduction
- Managed a large scale audition for **Antragni Idols**,which had a participation of **120+ individuals** from different colleges

Extra-Curricular Activities

Academic	-Achieved 4th position for creative writing at Regional Cultural and Literary Meet , also had a privilage to respresent Hyderabad Region at National Cultural and Literary Meet at New Delhi
Cultural	-Auditioned for Sa Re Ga Ma Pa Season 9 , a kannada singing reality show, showcasing vocal talent -1st position in vocabulary bee competition conducted by North South Foundation
Sports	-Participated in multiple District Level volleyball tournaments and awarded as Emerging Player in IVL -Won Gold in intra-hall, Silver in Inferno'23 and a Bronze in IVL'23 volleyball tournaments